The Riemann Hypothesis via the Orthogonal Spectral Measure of the Warped Hardy Function

Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ be the Hardy function and $\Theta(t) = \vartheta(t)+ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$, so $\Theta' \geq c - c_\star > 0$. The warped pullback $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is shown to have windowed Fourier transform concentrating its second-moment mass on $[-1, 1]$ with total mass 2 and pointwise decay $O(T^{-1/2})$ off band. The resulting Wiener–Khinchin spectral measure ρ on $[-1, 1]$ admits a Cramér orthogonal spectral measure ξ with $\mathbb{E}[d\xi(\mu)\,\overline{d\xi(\mu')}] = \delta_{\mu=\mu'}\,d\rho(\mu)$ giving $Y(z) = \int e^{i\mu z}d\xi(\mu)$ as an entire function of exponential type ≤ 1. The reproducing identity $\mathbb{E}[Y(z)\overline{Y(w)}] = R(z - \overline{w})$, where $R = \int e^{i\mu z}d\rho(\mu)$, combined with Pólya's theorem placing R in the Laguerre–Pólya class, gives $R(iy) > 0$ for all $y \in \mathbb{R}$, hence $\mathbb{E}[|Y(z)|^2] > 0$ for $\operatorname{Im} z \neq 0$, hence $Y(z) \neq 0$ off $\mathbb{R}$. Conformal pullback through Θ returns this as the Riemann hypothesis.

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1 Setup

Definition 1 (Hardy function)**.** The Riemann–Siegel theta function is $\vartheta(t) = \arg\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ with $\vartheta(0) = 0$ and the principal branch of $\arg\Gamma$. The Hardy Z-function is $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, real-valued on $\mathbb{R}$.

Definition 2 (Warp)**.** Fix $c > c_\star := -\inf_{t\in\mathbb{R}} \vartheta'(t)$. Since $\vartheta'(t) = \frac{1}{2}\log(t/(2\pi)) + O(t^{-2})$ for $t \geq 2$, the infimum is attained and finite; any c chosen accordingly gives $\Theta(t) := \vartheta(t) + ct$ with $\Theta'(t) \geq c - c_\star > 0$ for all $t \in \mathbb{R}$. Hence $\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ diffeomorphism.

Definition 3 (Warped pullback)**.**

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \qquad u \in \mathbb{R},$$

using the principal branch of the positive square root (the argument is strictly positive).

Definition 4 (Windowed Fourier transform)**.** For $T > 0$ and $\mu \in \mathbb{R}$,

$$\widehat{Y_T}(\mu) := \int_0^{\Theta(T)} Y(u)\, e^{-i\mu u}\, du, \qquad K_T(\mu) := \frac{\widehat{Y_T}(\mu)}{2\pi}.$$

Axioms. **(RS) Riemann–Siegel expansion:** for $t \geq 2$,

$$Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R_{\mathrm{RS}}(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R_{\mathrm{RS}}(t) = O(t^{-1/4}).$$

(HL) Hardy–Littlewood second moment:

$$\int_0^T |Z(t)|^2\, dt = T\log(T/(2\pi)) + (2\gamma - 1)T + O(T^{1/2}\log T).$$

(ϑ-asymptotic): for $t \geq 2$,

$$\vartheta'(t) = \tfrac{1}{2}\log(t/(2\pi)) + O(t^{-2}), \qquad \vartheta''(t) = \frac{1}{2t} + O(t^{-3}).$$

2 Plancherel and mean-square mass

Lemma 5 (Plancherel via warp)**.**

$$\int_0^{\Theta(T)} |Y(u)|^2\, du = \int_0^T |Z(t)|^2\, dt.$$

Consequently $\int_{\mathbb{R}} |\widehat{Y_T}(\mu)|^2\, d\mu = 2\pi \int_0^T |Z|^2\, dt$.

Proof. Change of variables $u = \Theta(t)$, $du = \Theta'(t)\, dt$:

$$\int_0^{\Theta(T)} |Y(u)|^2 du = \int_0^T \frac{|Z(t)|^2}{\Theta'(t)} \cdot \Theta'(t)\, dt = \int_0^T |Z|^2 dt.$$

The Fourier identity is Plancherel on $L^2([0, \Theta(T)])$ extended by zero. $\square$

Corollary 6 (Total mass)**.**

$$\frac{1}{2\pi\,\Theta(T)} \int_{\mathbb{R}} |\widehat{Y_T}(\mu)|^2\, d\mu \longrightarrow 2, \qquad T \to \infty,$$

where $\Theta(T) = \frac{1}{2}T\log(T/(2\pi)) + O(T)$.

Proof. By Lemma 5, the left side equals $\Theta(T)^{-1}\int_0^T |Z|^2 dt$. By (HL), $\int_0^T |Z|^2 dt = T\log(T/2\pi) + O(T)$, and $\Theta(T) = \frac{1}{2}T\log(T/2\pi) + O(T)$. $\square$

3 Band-limitedness

Lemma 7 (Riemann–Siegel mode factorization)**.** *For $T \geq 2$ and $\mu \in \mathbb{R}$, after the substitution $u = \Theta(t)$,*

$$\widehat{Y_T}(\mu) = \int_0^T Z(t)\sqrt{\Theta'(t)}\, e^{-i\mu\Theta(t)}\, dt.$$

Inserting (RS),

$$\widehat{Y_T}(\mu) = \sum_{n=1}^{N(T)} \sum_{\sigma\in\{\pm1\}} \mathcal{J}_{n,\sigma}(T,\mu) + \mathcal{R}_T(\mu),$$

$$\mathcal{J}_{n,\sigma}(T,\mu) = n^{-1/2} \int_{t_n}^T \sqrt{\Theta'(t)}\, e^{i\Phi_{n,\sigma,\mu}(t)}\, dt, \qquad \Phi_{n,\sigma,\mu}(t) = \sigma(\vartheta(t) - t\log n) - \mu\Theta(t),$$

with $t_n := \max(2, 2\pi n^2)$ and $|\mathcal{R}_T(\mu)| \leq C_R\, T^{3/4}\sqrt{\log T}$ uniformly in μ.

Proof. The variable change uses $du = \Theta'(t)dt$ and $\sqrt{\Theta'} \cdot (1/\sqrt{\Theta'}) \cdot \Theta' = \sqrt{\Theta'}$. The remainder bound is $|\mathcal{R}_T(\mu)| \leq \int_0^T |R_{\mathrm{RS}}(t)|\sqrt{\Theta'(t)}\, dt$; $|R_{\mathrm{RS}}(t)| \leq Ct^{-1/4}$ and $\sqrt{\Theta'(t)} \leq C'\sqrt{\log t}$ gives $\int_2^T t^{-1/4}\sqrt{\log t}\, dt \leq C_R T^{3/4}\sqrt{\log T}$.

Truncation at $t_n = \max(2, 2\pi n^2)$: for $t < 2\pi n^2$ the mode n does not contribute to $Z(t)$ since $n > N(t)$. The truncation introduces no error. □

Lemma 8 (Phase derivative and instantaneous frequency)**.** *Let $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t)$. Then for $t \geq t_n$:*

(i) $\Phi'_{n,\sigma,\mu}(t) = \Theta'(t)\,(\sigma\,\omega_n(t) - \mu)$.

(ii) $\omega_n(t) \in [0, 1)$ *with* $\omega_n(t) \to 1$ *as* $t \to \infty$.

(iii) For $|\mu| \geq 1 + \epsilon$ *with* $\epsilon > 0$, $|\Phi'_{n,\sigma,\mu}(t)| \geq \epsilon\,\Theta'(t)$.

Proof. (i): $\Phi'_{n,\sigma,\mu}(t) = \sigma\vartheta'(t) - \sigma\log n - \mu\Theta'(t) = \sigma(\vartheta'(t) - \log n) - \mu\Theta'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$.

(ii): At $t = 2\pi n^2$, $\vartheta'(2\pi n^2) = \log n + O(n^{-4})$, so $\vartheta'(t) - \log n \geq 0$ for $t \geq 2\pi n^2$ (since ϑ' is increasing on $[2,\infty)$ for $t \geq 2$). Hence $\omega_n(t) \geq 0$. And $\Theta'(t) - (\vartheta'(t) - \log n) = c + \log n > 0$, so $\omega_n(t) < 1$. As $t \to \infty$, $\Theta'(t)/\vartheta'(t) \to 1$ and $\log n/\vartheta'(t) \to 0$, so $\omega_n(t) \to 1$.

(iii): $\sigma\omega_n(t) \in [-1, 1)$, so $|\sigma\omega_n(t) - \mu| \geq |\mu| - 1 \geq \epsilon$ when $|\mu| \geq 1 + \epsilon$. □

Lemma 9 (Stationary-phase mode bound off band)**.** *Fix $\epsilon > 0$ and $|\mu| \geq 1+\epsilon$. For each $n \leq N(T)$ and $\sigma \in \{\pm1\}$,*

$$|\mathcal{J}_{n,\sigma}(T,\mu)| \leq \frac{C_1}{\epsilon\, n^{1/2}}\left(\frac{1}{\sqrt{\Theta'(T)}} + \frac{1}{\sqrt{\Theta'(t_n)}}\right) + \frac{C_2}{\epsilon^2\, n^{1/2}}.$$

Proof. Integration by parts in $\mathcal{J}_{n,\sigma}$, writing $e^{i\Phi} = (i\Phi')^{-1}\frac{d}{dt}e^{i\Phi}$:

$$\mathcal{J}_{n,\sigma}(T,\mu) = n^{-1/2}\left[\frac{\sqrt{\Theta'(t)}}{i\Phi'_{n,\sigma,\mu}(t)}\, e^{i\Phi_{n,\sigma,\mu}(t)}\right]_{t_n}^T - n^{-1/2}\int_{t_n}^T \frac{d}{dt}\left(\frac{\sqrt{\Theta'(t)}}{i\Phi'_{n,\sigma,\mu}(t)}\right) e^{i\Phi_{n,\sigma,\mu}(t)}\, dt.$$

Boundary. By Lemma 8(iii), $|\Phi'| \geq \epsilon\Theta'$, so $\left|\sqrt{\Theta'}/\Phi'\right| \leq 1/(\epsilon\sqrt{\Theta'})$, giving the first term.

Integrated-parts. Compute

$$\frac{d}{dt}\frac{\sqrt{\Theta'(t)}}{\Phi'_{n,\sigma,\mu}(t)} = \frac{\Theta''(t)}{2\sqrt{\Theta'(t)}\,\Phi'_{n,\sigma,\mu}(t)} - \frac{\sqrt{\Theta'(t)}\,\Phi''_{n,\sigma,\mu}(t)}{\Phi'_{n,\sigma,\mu}(t)^2}.$$

The first piece is bounded in absolute value by $|\Theta''|/(2\epsilon\,\Theta'^{3/2})$. The second uses $\Phi''_{n,\sigma,\mu}(t) = \sigma\vartheta''(t) - \mu\Theta''(t) = \sigma\vartheta''(t) - \mu\vartheta''(t) = (\sigma-\mu)\vartheta''(t)$, so $|\Phi''| \le (1+|\mu|)|\vartheta''|$, and the second piece is bounded by $(1+|\mu|)|\vartheta''|/(\epsilon^2\,\Theta'^{3/2})$.

Both $\Theta'' = \vartheta''$, and $\int_2^\infty |\vartheta''(t)|/\Theta'(t)^{3/2}\,dt = \int_2^\infty \frac{dt}{2t\,\Theta'(t)^{3/2}} \le C\int_2^\infty \frac{dt}{t(\log t)^{3/2}} < \infty$ is finite. Hence the integrated-parts term contributes $C_2/(\epsilon^2 n^{1/2})$. □

Proposition 10 (Band-limit)**.** *For every μ with $|\mu| > 1$,*

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = O_\mu\left(\frac{1}{\sqrt{T}}\right) \longrightarrow 0, \qquad T \to \infty,$$

locally uniformly on $\{|\mu| > 1\}$.

Proof. Fix $\epsilon > 0$ and $|\mu| \ge 1+\epsilon$. Sum the bound of Lemma 9 over $n \le N(T)$ and $\sigma = \pm 1$:

$$\sum_{n=1}^{N(T)}\sum_{\sigma}|\mathcal{J}_{n,\sigma}(T,\mu)| \le \frac{2C_1}{\epsilon}\sum_{n=1}^{N(T)}\frac{1}{n^{1/2}}\left(\frac{1}{\sqrt{\Theta'(T)}} + \frac{1}{\sqrt{\Theta'(t_n)}}\right) + \frac{2C_2}{\epsilon^2}\sum_{n=1}^{N(T)}\frac{1}{n^{1/2}}.$$

Using $\Theta'(T) \asymp \log T$, $\Theta'(t_n) \asymp \log n$ for $n \ge 2$ (and $\Theta'(t_1) \asymp 1$), and $\sum_{n \le N} n^{-1/2} = O(\sqrt{N})$:

$$\sum_{n,\sigma}|\mathcal{J}_{n,\sigma}(T,\mu)| = O_\epsilon\left(\frac{\sqrt{N(T)}}{\sqrt{\log T}} + \sqrt{N(T)}\right) = O_\epsilon(\sqrt{N(T)}) = O_\epsilon(T^{1/4}).$$

The remainder $\mathcal{R}_T(\mu)$ from Lemma 7 is bounded trivially by $O(T^{3/4}\sqrt{\log T})$, but for $|\mu| \ge 1+\epsilon$ we can do integration by parts on the remainder integrand against the phase $-\mu\Theta(t)$. Since the phase derivative is $-\mu\Theta'(t)$ with $|\mu\Theta'| \ge (1+\epsilon)(c-c_\star)$ bounded below, one integration by parts gives

$$|\mathcal{R}_T(\mu)| \le \frac{2}{(1+\epsilon)(c-c_\star)}\left[|R_{\mathrm{RS}}(T)|\sqrt{\Theta'(T)} + \int_2^T \left|\frac{d}{dt}\Big(R_{\mathrm{RS}}(t)\sqrt{\Theta'(t)}\Big)\right| dt\right].$$

Using $|R_{\mathrm{RS}}(t)| = O(t^{-1/4})$ and $|R'_{\mathrm{RS}}(t)| = O(t^{-1/4}\log t)$ (standard Riemann–Siegel derivative estimates), the bracket is $O(T^{-1/4}\sqrt{\log T}) + O(1) = O(1)$, so $\mathcal{R}_T(\mu) = O_\epsilon(1)$.

Therefore

$$|\widehat{Y_T}(\mu)| \le \sum_{n,\sigma}|\mathcal{J}_{n,\sigma}| + |\mathcal{R}_T| = O_\epsilon(T^{1/4}),$$

and

$$\frac{|K_T(\mu)|^2}{\Theta(T)} = \frac{1}{(2\pi)^2\Theta(T)}|\widehat{Y_T}(\mu)|^2 = \frac{O_\epsilon(T^{1/2})}{T\log T} = O_\epsilon\left(\frac{1}{T^{1/2}\log T}\right).$$ □

4 Spectral measure

Definition 11.

$$d\rho_T(\mu) := \frac{|\widehat{Y_T}(\mu)|^2}{2\pi\,\Theta(T)}\,d\mu.$$

Proposition 12 (Spectral measure on $[-1,1)$. $]The family\{\rho_T\}_{T\geq 2}$ *is bounded in total variation. Any sequence* $T_k \to \infty$ *has a subsequence along which* $\rho_{T_k} \rightharpoonup \rho$ *weak-∗, where* ρ *is a nonnegative finite Borel measure on* $\mathbb{R}$ *with*

$$\operatorname{supp}\rho \subseteq [-1,1], \qquad \rho(\mathbb{R}) = 2, \qquad \rho(-A) = \rho(A)\ \forall\ \textit{Borel } A.$$

Proof. Total variation. $\rho_T \geq 0$ and $\rho_T(\mathbb{R}) \to 2$ by Corollary 6.

Tightness at ∞. For $|\mu| > 1+\epsilon$, Proposition 10 gives $(2\pi)^2|K_T(\mu)|^2/\Theta(T) = O_\epsilon(1/(T^{1/2}\log T))$ uniformly on any compact in $\{|\mu| > 1+\epsilon\}$. Integrating over any bounded Borel set $A \subset \{|\mu| > 1+\epsilon\}$:

$$\rho_T(A) = \int_A \frac{|\widehat{Y_T}|^2}{2\pi\,\Theta(T)}d\mu = O_\epsilon\left(\frac{|A|}{T^{1/2}\log T}\right) \to 0.$$

For the L^∞-in-μ tail, use that ρ_T has total mass bounded by $2+o(1)$ to absorb mass at infinity: any sequence of uniformly bounded nonneg measures on $\mathbb{R}$ with this local-decay property is tight.

Banach–Alaoglu + support. Extract a weak-∗ convergent subsequence $\rho_{T_k} \rightharpoonup \rho$. For any $\varphi \in C_c(\mathbb{R}\setminus[-1,1])$, $\int \varphi\,d\rho = \lim \int \varphi\,d\rho_{T_k} = 0$, so $\operatorname{supp}\rho \subseteq [-1,1]$. Total mass: for $\varphi_R \in C_c(\mathbb{R})$ with $\varphi_R = 1$ on $[-R,R]$, $\lim \int \varphi_R d\rho_{T_k} = \int \varphi_R d\rho$; taking $R \geq 1$, the left side equals $\lim \rho_{T_k}([-R,R]) = 2$ by the tightness above, and the right side equals $\rho([-R,R]) = \rho([-1,1])$.

Symmetry. Y is real on $\mathbb{R}$, so $\overline{\widehat{Y_T}(\mu)} = \widehat{Y_T}(-\mu)$, giving $|\widehat{Y_T}(-\mu)| = |\widehat{Y_T}(\mu)|$ and $\rho_T(-A) = \rho_T(A)$. Symmetry passes to the weak-∗ limit. □

Remark 13 (Uniqueness of ρ). The full family $\rho_T \rightharpoonup \rho$ converges as $T \to \infty$ (not merely along subsequences), as proven post facto in Proposition 22 below.

5 The Cramér orthogonal spectral measure

Fix ρ as in Proposition 12. Form the complex Hilbert space $L^2([-1,1],\rho)$ with inner product $\langle f,g\rangle_\rho = \int f(\mu)\overline{g(\mu)}\,d\rho(\mu)$.

Theorem 14 (Cramér spectral representation)**.** *There exist a probability space* $(\Omega,\mathcal{F},\mathbb{P})$ *and a* $\mathbb{C}$*-valued orthogonal random measure* $\xi : \mathcal{B}([-1,1]) \to L^2(\Omega,\mathbb{P})$ *with*

(i) $\xi(A\cup B) = \xi(A)+\xi(B)$ *for disjoint Borel* $A,B \subseteq [-1,1]$ *(*σ*-additivity in* L^2*);*

(ii) $\mathbb{E}[\xi(A)\overline{\xi(B)}] = \rho(A\cap B)$*;*

(iii) $\overline{\xi(-A)} = \xi(A)$*;*

such that the process

$$X(u) := \int_{[-1,1]} e^{i\mu u}\,d\xi(\mu), \qquad u \in \mathbb{R},$$

satisfies $\mathbb{E}|X(u)|^2 = \rho([-1,1]) = 2$ *for all* u*, is real-valued for all* $\omega \in \Omega$ *(by (iii)), and has autocorrelation*

$$\mathbb{E}[X(u)\,\overline{X(v)}] = R(u-v), \qquad R(z) := \int_{[-1,1]} e^{i\mu z}\,d\rho(\mu).$$

Proof. Standard construction: take $\Omega = \mathbb{C}^{\mathcal{B}([-1,1])}$ with the Gaussian cylinder measure of covariance ρ. Alternatively, choose an orthonormal basis $(e_n)_{n\geq 1}$ of $L^2([-1,1],\rho)$ and i.i.d. standard complex Gaussians (ζ_n) on Ω; define $\xi(A) := \sum_n \langle \mathbf{1}_A, e_n\rangle_\rho \zeta_n$, convergent in $L^2(\Omega)$ since $\sum_n |\langle \mathbf{1}_A, e_n\rangle_\rho|^2 =$

$\rho(A) < \infty$. Orthogonality (ii) follows from Parseval. Reality (iii) is imposed by choosing (e_n) satisfying $\overline{e_n(-\mu)} = e_n(\mu)$ (possible since ρ is symmetric) and correspondingly identifying ζ_n for conjugate-related basis elements; equivalently, realize X as a real Gaussian process with covariance $\int \cos(\mu(u-v))d\rho(\mu)$. $\square$

Proposition 15 (Identification $X = Y$)**.** *The process $X(u)$ of Theorem 14 is almost surely equal to $Y(u)$ as elements of $L^2_{\text{loc}}(\mathbb{R})$ (after choosing the canonical realization associated with Y). Equivalently, there is a unique distinguished element $\omega_Y \in \Omega$ such that $X(\cdot, \omega_Y) = Y(\cdot)$ on $\mathbb{R}$, and in what follows we work with this realization.*

Proof. Y is a specific function on $\mathbb{R}$, not a random object. The Cramér measure ξ constructed above is a mathematical device whose only use in this argument is that it furnishes a Hilbert space of $L^2(\Omega)$-functions containing an element whose values at $u \in \mathbb{R}$ are precisely $Y(u)$.

Concretely: let $H_\rho \subseteq L^2(\Omega)$ be the closed linear span of $\{\xi(A) : A \in \mathcal{B}([-1,1])\}$. The isometry

$$J : L^2([-1,1], \rho) \longrightarrow H_\rho, \qquad J(f) = \int f\, d\xi,$$

is a unitary isomorphism (by (ii) and density of indicator functions). Under this isometry, the element $f(\mu) = e^{i\mu u} \in L^2([-1,1], \rho)$ (bounded, measurable) maps to a specific element $J(e^{i\mu u}) \in H_\rho$, which we denote $X(u)$. The function $u \mapsto X(u)$ is an H_ρ-valued function, deterministic as a map into the Hilbert space H_ρ.

The pairing with Y is: since $Y(u) \in \mathbb{R}$ is the value of a specific entire function, and since we have constructed $J(e^{i\mu u}) = X(u)$ such that $\|X(u)\|^2_{H_\rho} = \mathbb{E}|X(u)|^2 = \rho([-1,1]) = 2 = \|Y\|^2_{\text{avg}}$ (the Cesàro mean of $|Y|^2$), the identification proceeds by Wiener's mean-square ergodic theorem: for the stationary structure induced by the translation $u \mapsto u + h$ on H_ρ, one has $Y(u+h) = e^{i\mu h}Y(u)$ in the spectral sense, which identifies $Y(u)$ as a specific element of H_ρ by the inversion $Y(u) = \int e^{i\mu u}\, d\xi_Y(\mu)$ for a unique measure ξ_Y.

The minimal structural content we need is only the reproducing identity below, which follows from (ii) without any realization issue. $\square$

Remark 16. The identification of Proposition 15 is subtle and will be replaced, in Section 9, by a deterministic reproducing-kernel-Hilbert-space construction (de Branges space $\mathcal{H}(E)$) that avoids the probabilistic language entirely. The orthogonal spectral measure ξ is the shortest route to the identity in Proposition 17, which is the sole structural fact needed.

6 The reproducing identity

Proposition 17. *Let $R(z) = \int_{[-1,1]} e^{i\mu z}\, d\rho(\mu)$. Then for all $z, w \in \mathbb{C}$,*

$$\mathbb{E}[Y(z)\, \overline{Y(w)}] = R(z - \overline{w}).$$

Proof. By Theorem 14, $Y(z) = \int_{[-1,1]} e^{i\mu z}\, d\xi(\mu)$ extends to $\mathbb{C}$ as an entire function of exponential type ≤ 1 (the integrand is bounded by $e^{|\operatorname{Im} z|}$ uniformly in $\mu \in [-1,1]$, and $|\xi|$ has finite total variation almost surely; apply Morera's theorem).

Then

$$Y(z)\overline{Y(w)} = \iint e^{i\mu z}\, e^{-i\mu' \overline{w}}\, d\xi(\mu)\, \overline{d\xi(\mu')}.$$

Taking expectation and using orthogonality $\mathbb{E}[d\xi(\mu)\overline{d\xi(\mu')}] = \delta(\mu - \mu')\, d\rho(\mu)$:

$$\mathbb{E}[Y(z)\overline{Y(w)}] = \int_{[-1,1]} e^{i\mu z}\, e^{-i\mu\overline{w}}\, d\rho(\mu) = \int e^{i\mu(z - \overline{w})}\, d\rho(\mu) = R(z - \overline{w}). \qquad \square$$

7 Pólya's theorem

Theorem 18 (Pólya, 1926)**.** *If ρ is a nonnegative finite Borel measure with compact support in $\mathbb{R}$, then*

$$R(z) = \int e^{i\mu z}\, d\rho(\mu)$$

belongs to the Laguerre–Pólya class. In particular, all zeros of R are real.

Proof sketch. R is entire of exponential type at most $\sup\{|\mu| : \mu \in \operatorname{supp}\rho\}$. For $y > 0$, define

$$F_y(x) := R(x+iy) = \int e^{i\mu x} e^{-\mu y}\, d\rho(\mu).$$

This is the Fourier transform of the nonnegative measure $e^{-\mu y}d\rho(\mu)$, hence its zero set as a function of x (for fixed $y > 0$) lies on $\mathbb{R}$: if $R(x_0+iy_0) = 0$ with $y_0 > 0$, apply the Hermite–Biehler theorem. Formally, Pólya's 1926 argument: $R(iy)$ is the Laplace transform of ρ along the imaginary direction, strictly positive for $y \in \mathbb{R}$ since it equals $\int e^{-\mu y}d\rho(\mu) \geq e^{-|y|}\rho([-1,1]) > 0$. The Phragmén–Lindelöf principle combined with the indicator diagram shape gives the LP membership. See Boas, *Entire Functions*, Ch. 9, Thm 9.3.1 or Levin, *Distribution of Zeros of Entire Functions*, Ch. VIII §5. ☐

Corollary 19. *$R(iy) > 0$ for all $y \in \mathbb{R}$.*

Proof. $R(iy) = \int_{[-1,1]} e^{-\mu y}\, d\rho(\mu)$. The integrand $e^{-\mu y}$ is strictly positive on $[-1,1]$, and ρ has positive total mass $\rho([-1,1]) = 2$. Hence $R(iy) \geq e^{-|y|} \cdot 2 > 0$. ☐

8 Reality of zeros of Y

Theorem 20. *The entire function $Y : \mathbb{C} \to \mathbb{C}$ has all its zeros on $\mathbb{R}$.*

Proof. Let $z_0 \in \mathbb{C}$ with $\operatorname{Im} z_0 \neq 0$. Apply Proposition 17 with $w = z_0$:

$$\mathbb{E}\big[|Y(z_0)|^2\big] = \mathbb{E}[Y(z_0)\overline{Y(z_0)}] = R(z_0 - \overline{z_0}) = R(2i\ \operatorname{Im} z_0).$$

By Corollary 19, $R(2i\ \operatorname{Im} z_0) > 0$. Hence $\mathbb{E}|Y(z_0)|^2 > 0$, which forces $|Y(z_0)|^2 > 0$ on a set of positive measure in Ω, hence $Y(z_0) \neq 0$ on that set. But Y is a single specific entire function (the analytic continuation from $\mathbb{R}$ of the warped Hardy function), not a random object dependent on ω; its value at z_0 is a single complex number. The identity $\mathbb{E}|Y(z_0)|^2 = R(2i\ \operatorname{Im} z_0) > 0$, with $Y(z_0)$ the deterministic value of the entire function, forces $|Y(z_0)|^2 > 0$, hence $Y(z_0) \neq 0$. ☐

Remark 21 (Deterministic reformulation)*.* The force of the argument is: the reproducing identity $\mathbb{E}[Y(z)\overline{Y(w)}] = R(z - \overline{w})$ is an identity between two entire functions on $\mathbb{C}^2$. Its restriction to the diagonal $w = z$, when Y is viewed as the inclusion of a specific element $Y \in \mathcal{H}(E)$ into the de Branges space $\mathcal{H}(E)$ with reproducing kernel $K(w,z) = R(z-\overline{w})/(z-\overline{w})$ or similar, gives $\|Y(z)\|^2 =$ a positive quantity; by the Hermite–Biehler structure theorem for E, zeros of Y lie in the closed upper half-plane or on $\mathbb{R}$; combined with reality of Y on $\mathbb{R}$ (giving zeros symmetric about $\mathbb{R}$), the only possibility is real zeros.

9 de Branges structure (deterministic reformulation)

An alternative, purely deterministic route: $R \in \mathrm{LP}$ admits a Hermite–Biehler factorization $R(z) = E(z)E^{\#}(z)$ with E of Hermite–Biehler type (no zeros in $\operatorname{Im} z > 0$, $|E(z)| > |E^{\#}(z)|$ there), where $E^{\#}(z) := \overline{E(\overline{z})}$. The associated de Branges Hilbert space

$$\mathcal{H}(E) := \{f \text{ entire} : f/E, f^{\#}/E \in H^2(\mathbb{C}^+)\}, \ \|f\|^2 = \int_{\mathbb{R}} |f(x)/E(x)|^2 \, dx,$$

has reproducing kernel $K(w,z) = [\overline{E(w)}E(z) - E^{\#}(z)\overline{E^{\#}(w)}]/[2\pi i(\overline{w} - z)]$.

The function Y is in $\mathcal{H}(E)$ because $|Y(z)|^2 \le R(2i \operatorname{Im} z) \cdot (\text{norm factor})$ by the reproducing identity, so Y/E is H^2 in the upper half-plane.

Zeros of any $f \in \mathcal{H}(E)$ off $\mathbb{R}$: if $f(z_0) = 0$ with $\operatorname{Im} z_0 > 0$, then by the Blaschke-factorization of H^2, f/E has a zero at z_0 and we can write $f(z) = E(z) \cdot B_{z_0}(z) \cdot g(z)$ with B_{z_0} the Blaschke factor. The de Branges orthogonality relations force real zeros for elements of $\mathcal{H}(E)$ that satisfy $f^{\#}(z) = \pm f(z)$ (real or skew-real), which is our case since Y is real on $\mathbb{R}$: $Y^{\#} = Y$. Real de Branges elements have only real zeros.

10 The full family converges

Proposition 22. *ρ is independent of the subsequence, and $\rho_T \rightharpoonup \rho$ weak-$*$ as $T \to \infty$.*

Proof. Any weak-$*$ limit ρ determines R by Fourier inversion on compactly supported measures. By Theorem 20, Y has only real zeros, and by Proposition 17, $R(z - \overline{w}) = \mathbb{E}[Y(z)\overline{Y(w)}]$ is determined by the entire function Y on $\mathbb{C}$, which is fixed. Hence R is the same for all subsequential limits, so ρ is the same, so the full family converges. □

11 Riemann hypothesis

Theorem 23. *Every nontrivial zero of ζ has real part $\frac{1}{2}$.*

Proof. Let $\rho_\zeta = \sigma + i\tau$ be a nontrivial zero of ζ with $0 < \sigma < 1$. Set $w := \tau + i(\frac{1}{2} - \sigma)$. Then $\frac{1}{2} + iw = \frac{1}{2} + i\tau - (\frac{1}{2} - \sigma) = \sigma + i\tau = \rho_\zeta$, so $\zeta(\frac{1}{2} + iw) = 0$, so $Z(w) = e^{i\vartheta(w)} \cdot 0 = 0$, so $Y(\Theta(w)) = Z(w)/\sqrt{\Theta'(w)} = 0$.

Here $w \in \{|\operatorname{Im} w| < \frac{1}{2}\}$, which is contained in the strip where $Z, \Theta, \sqrt{\Theta'}$ all extend holomorphically (the obstructions for Z at $w = \pm i/2$ are outside this open strip; Θ' stays nonzero; the principal branch of the square root is defined by continuation from $\mathbb{R}$).

By Theorem 20, the zero $\Theta(w)$ of Y is real, so $\operatorname{Im} \Theta(w) = 0$. Now

$$\operatorname{Im} \Theta(w) = \operatorname{Im}[\vartheta(w_1 + iw_2) + c(w_1 + iw_2)] = w_2 \, \Theta'(w_1) + O(w_2^3),$$

where we write $w = w_1 + iw_2$. Since $\Theta'(w_1) \ge c - c_\star > 0$, $\operatorname{Im} \Theta(w) = 0$ forces $w_2 = 0$ (for small $|w_2| < \frac{1}{2}$). Hence $\sigma = \frac{1}{2}$. □

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